Vadim Egorov

Software Engineer | Web Development & AI Research

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PROFILE

Software Engineer with extensive experience in web development and implementation of complex business logic within large-scale applications. Skilled in building scalable frontend architectures, structured state management systems, and integrating backend services to support real-world operational workflows.

Currently pursuing an M.S. in Artificial Intelligence, Data Science, and Engineering, with research focused on EEG signal modeling and self-supervised learning approaches such as Joint Embedding Predictive Architecture (JEPA) to improve representation learning from neural data.

PROFESSIONAL EXPERIENCE

Graduate Student Assistant, *UAA College of Engineering* — Jan 2026 – Present, Anchorage, AK

- Provided administrative and operational support to the College of Engineering Dean's Office, assisting students, faculty, and external stakeholders.
- Coordinated and supported college events, managing logistics, and execution.
- Contributed to internal administrative initiatives and special projects essential to the college's daily operations.

Software Developer, *Flyntlok, Inc.* — Apr 2023 – Dec 2025, Anchorage, AK

- Developed and maintained core features of a cloud-based DMS platform for equipment dealerships, supporting inventory, rentals, work orders, and sales workflow.
- **Led development of the Sales Project module**, a centralized system for managing quotes, cost worksheets, part and machine purchase orders, and multi-stage sales approvals — enabling dealers to structure complex deals, track profitability, and streamline sales-to-operations handoffs.
- Built scalable frontend systems using **React and TypeScript**, leveraging **Redux** for structured state management across complex, data-heavy interfaces.
- Optimized backend logic and SQL queries to improve reporting, auditing, and real-time pricing performance.
- Collaborated cross-functionally with product, QA, and backend teams to deliver reliable full-stack solutions aligned with dealer-specific business processes.

eLearning Technology Student Technician, *UAA College of Engineering* — Aug 2021 – Dec 2022, Anchorage, AK

- Provided frontline technical support to students, faculty, and staff within the College of Engineering at the University of Alaska Anchorage, ensuring reliable operation of eLearning technologies and classroom systems.
- Monitored and tested audio/video infrastructure, assisted users with classroom connectivity, recorded lectures, and diagnosed and resolved real-time technical issues to minimize disruptions.

Student Assistant, *UAA Summer Engineering Academies* — May 2022 – Aug 2022, Anchorage, AK

- Executed lead teacher lesson plans while maintaining structured daily classroom schedules to ensure a consistent learning environment.
- Provided individualized academic support to students.
- Fostered students' social, emotional, and intellectual development through attentive supervision and engagement.

PERSONAL PROJECTS

PomoFriends , *Social Productivity Platform*
Next.js, Firebase, TypeScript, Tailwind CSS

Developed a productivity web app inspired by the Pomodoro Technique, combining time management with social accountability features such as group sessions, messaging, and real-time status updates. Implemented Firebase Authentication and Realtime Database to manage users, groups, and synchronized state across active sessions. Emphasized scalable architecture, clean UI/UX, and a responsive design using Next.js and Tailwind CSS.

sup4bubb4 , *Livestream Music Queue*
Svelte, Firebase, StreamElements API, Tailwind CSS

Built a web app that archives livestream replays alongside their corresponding music queues for a VTuber fan community. Integrated YouTube scraping and the StreamElements API to synchronize streams with song request data. Used Firebase for structured metadata storage and developed a responsive UI with Svelte and Tailwind CSS.

Predicting Student Learning Outcomes Using EEG
Individual Research

Independently designed and conducted a theoretical project exploring how neural synchronization between students and instructors could influence learning outcomes. Researched academic literature on EEG-based engagement and attention modeling. Simulated data analysis workflows.

Flower Classification Model
Python, TensorFlow, Keras, Jupyter

Developed and trained a CNN to classify 14 flower species, achieving over 92% test accuracy. Applied image preprocessing and augmentation, tuned hyperparameters with dropout and batch normalization, and evaluated performance using confusion matrices and Grad-CAM visualizations.

NoteBuds
GraphQL, React, TypeScript, PostgreSQL

Led development of a web-based note-taking and collaboration app. Designed a relational schema for note ownership, permissions, and version history. Built a strictly typed GraphQL API using Prisma and implemented a React frontend with markdown editing and search features. Emphasized clean architecture and modular design to support future real-time and offline capabilities.

EDUCATION

Master of Science in Artificial Intelligence, Data Science, and Engineering, Jan 2026 – Present
University of Alaska, Anchorage Anchorage, AK

Bachelor of Science in Computer Science, *University of Alaska Anchorage* Aug 2018 – Dec 2022
Anchorage, AK

SKILLS

Software Engineering — TypeScript | Python | React | Redux | GraphQL | REST | SQL | System Design | TDD

Machine Learning — TensorFlow | Keras | PyTorch | Model Training | Self-Supervised Learning | Data Preprocessing

Tools & Infrastructure — Git | Docker | Linux/Unix CLI | Chrome DevTools | VS Code | Copilot

Business & Domain Knowledge — Process automation for equipment dealership operations | Rental, service, sales, and inventory workflows | Work order and asset lifecycle management | Quoting, purchasing, pricing, and warranty systems

Soft Skills — Agile development (sprint planning, retrospectives) | Cross-functional collaboration (product, support, QA) | Clear and proactive communication | Problem-solving in production environments | Mentoring and documentation writing | Attention to edge cases and business logic details